AFRICAN LEADERSHIP UNIVERSITY
[BSE]
[ML TECHNIQUES II]
[SUMMATIVE ASSIGNMENT]

Reinforcement Learning Summative Assignment Report

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Video Recording: https://www.youtube.com/watch?v=Lh_TYrlzdoU
GitHub Repository: https://github.com/ChristianIshimwe7/Christian_Ishimwe_rl_summative

1. Project Overview

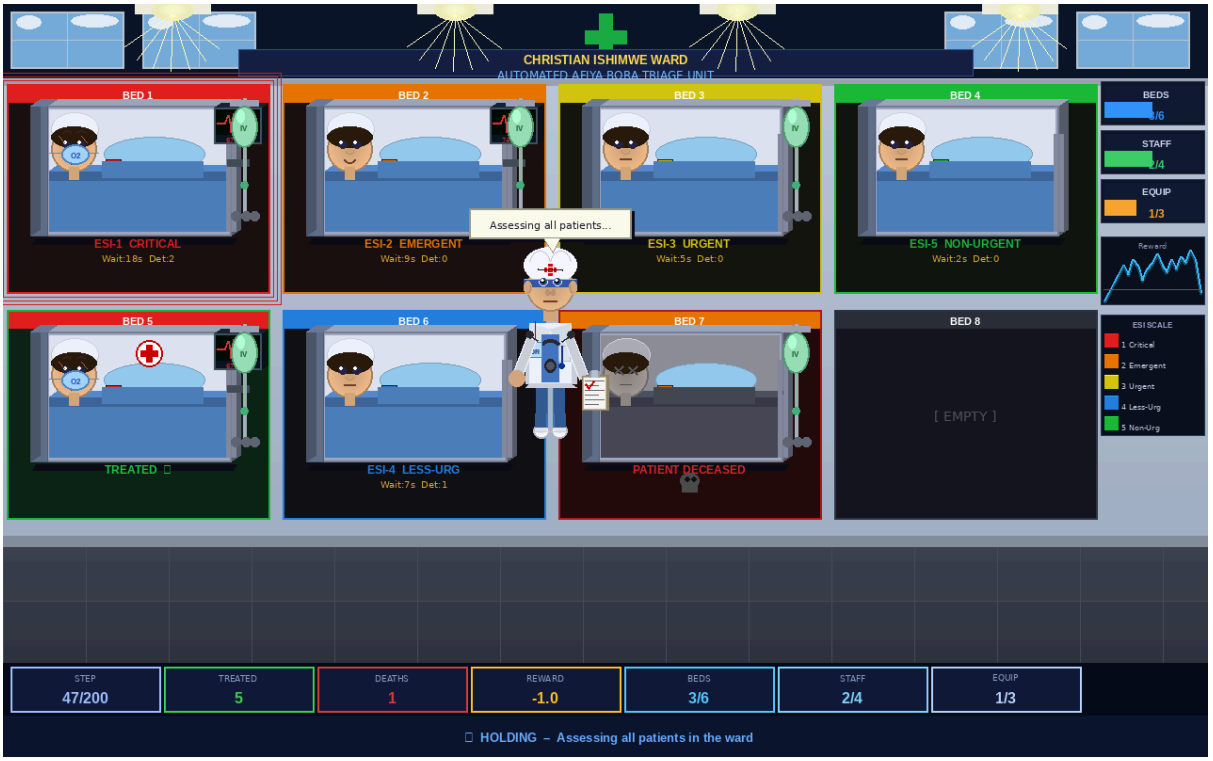


Fig 0: Project overview prototype image

This project implements a reinforcement learning environment simulating a hospital emergency triage unit, the Automated **Afiya Bora** Triage Unit. It addresses the challenge of constrained resource allocation under time pressure, requiring an AI agent to prioritize patients by severity while managing limited hospital resources. Patients deteriorate stochastically if untreated, with ESI-1 critical cases facing mortality when treatment is delayed. The environment was built as a custom Gymnasium setup with a 43-dimensional observation space and nine discrete actions. Four RL algorithms were benchmarked, with PPO achieving the highest mean reward (+58.2), converging fastest, and generalizing best to unseen triage scenarios.

2. Environment Description

a. Agent(s)

The agent represents an **AI-driven triage coordinator (Doctor or Nurse)** stationed at the central nursing desk of the emergency ward. At each time step, the agent observes the full clinical state of all eight beds, including

patient severity, wait time, treatment status, and vital deterioration count, as well as the current availability of all three resource pools. Based on this observation, the agent selects one of nine discrete actions: hold (do nothing) or treat one specific patient. The agent embodies the role of a senior triage nurse with the authority to allocate beds, staff, and equipment. Its goal is to minimize patient mortality while maximizing throughput, learning through trial and error which priority orderings and resource allocation strategies maximize long-term cumulative reward.

b. Action Space

The action space is **Discrete (9),** nine mutually exclusive actions per time-step:

Action ID	Description	Resource Cost	Condition
0	HOLD – do nothing this time-step	None	Any state
1	Treat Patient 1 (Bed 1)	ESI-1/2: 1 bed+2 staff+1 equip	Patient alive & untreated
2	Treat Patient 2 (Bed 2)	ESI-3: 1 bed+1 staff	Patient alive & untreated
3	Treat Patient 3 (Bed 3)	ESI-4/5: 1 bed only	Patient alive & untreated
4–8	Treat Patients 4–8 (Beds 4–8)	Same resource rules as above	Patient alive & untreated

Resource consumption scales with severity: ESI-1/2 patients require a bed, two staff members, and one equipment unit; ESI-3 requires a bed and one staff member; ESI-4/5 requires only a bed. If insufficient resources exist, the action is penalized rather than executed. This design forces the agent to learn resource-aware prioritization rather than treating the highest-severity patient indiscriminately.

c. Observation Space

The observation is a **Box (43)** float32 vector with all values normalized to [0, 1]:

Feature	Index	Range	Description
Patient Severity	0	[0, 1]	ESI level normalized by 5
Patient Wait Time	1	[0, 1]	Time-steps waiting, normalized by 200
Is Treated	2	{0, 1}	Binary: whether the patient has been treated
Is Alive	3	{0, 1}	Binary: patient alive status
Deterioration Count	4	[0, 1]	Number of deterioration events, norm. by 10
Beds Available	40	[0, 1]	Remaining beds normalized by MAX_BEDS (6)
Staff Available	41	[0, 1]	Remaining staff normalized by MAX_STAFF (4)

Feature	Index	Range	Description
Equipment Available	42	[0, 1]	Remaining equipment. norm. by MAX_EQUIPMENT (3)

The 43 features comprise a flattened 8×5 patient matrix (rows = patients, columns = severity, wait, treated, alive, deterioration) concatenated with three global resource scalars. All values are normalized to [0, 1] to stabilize neural network training. Patients with no active slot have all features set to zero, creating a clearly distinguishable state for empty beds.

d. Reward Structure

The reward function is **shaped** to guide the agent towards both correct prioritization and efficient resource usage:

Event	Reward	Rationale
Treat ESI-1 (Critical) patient	+10.0	Highest priority; immediate risk to life
Treat ESI-2/3 (Moderate) patient	+5.0	Significant urgency, moderate resource use
Treat ESI-4/5 (Minor) patient	+2.0	Low urgency; still rewarded for throughput
Correct priority order bonus	+2.0	Agent chose highest-severity untreated patient
Patient discharge (treated, t>10)	+3.0	Successful throughput; frees resources
Patient death (ESI-1 deteriorated)	-20.0	Catastrophic failure; maximum penalty
Wrong priority / dead patient	-3.0	Inefficient resource allocation penalized
Idle with critical patients	-1.0	Per untreated critical patient per time-step

The asymmetric structure, where a patient's death carries a 20 penalty versus +10 for a successful critical treatment, is intentional: it forces the agent to avoid catastrophic outcomes rather than simply maximizing throughput. The +2 bonus for correct priority ordering provides a dense training signal that accelerates early learning of the ESI triage protocol.

Terminal Conditions and Episode Structure

- Episode length: maximum 200 time-steps
- Truncation: step count reaches 200
- Termination: total deaths >= 3, or all patient slots are empty
- New patients arrive stochastically (15% chance per step) to maintain ward occupancy
- Treated patients discharged after 10+ wait time-steps, freeing resources for new admissions

3. System Analysis and Design

a. Deep Q-Network (DQN)

DQN was implemented using Stable Baselines 3 with an MLP policy. The network maps the 43-dimensional observation to Q-values for each of the 9 actions. Key features include: an **experience replay buffer** (50k–200k transitions) that decorrelates temporal training data; a **target network** updated at configurable intervals (tau = 0.01–1.0) to stabilize the Bellman targets; and an **epsilon-greedy exploration strategy** with linear annealing from 1.0 to a configurable final epsilon. The best-performing configuration (Run 7) used a [256×256] network, a 200k replay buffer, soft target updates (tau=0.01), and a moderate exploration fraction (0.30), achieving a mean episode reward of +55.8. DQN was well-suited to this environment due to its ability to learn stable Q-value estimates offline from replayed experience, which is particularly valuable when death events (−20 penalty) are rare but highly penalized.

b. Policy Gradient Method ([REINFORCE/PPO])

REINFORCE

REINFORCE was implemented via SB3's A2C class with the value function coefficient set to zero (vf_coef=0), reducing it to a pure Monte Carlo policy gradient method. The algorithm collects full episodes (n_steps = 200, matching maximum episode length) and updates the policy using discounted returns as the advantage estimate. No bootstrapping is performed, ensuring unbiased but high-variance gradient estimates. The best run (Run 7) used a [512×256] network and achieved +35.8 mean reward. REINFORCE was the weakest performer due to **high gradient variance**, particularly in episodes with rare death events. However, it served as an important baseline, demonstrating that policy gradient methods can learn the triage task even without a critic.

Proximal Policy Optimization (PPO)

PPO was implemented with SB3's default clip-range surrogate objective. The clipped probability ratio (epsilon = 0.2 for best runs) prevents overly large policy updates, substantially improving training stability over REINFORCE. PPO uses a shared actor-critic network, where the value function provides a baseline for advantage estimation via Generalized Advantage Estimation (GAE). The best configuration (Run 10) used a deep [256×256×128] network, n_steps=2048, batch size=64, 10 epochs per update, and entropy coefficient 0.01 for exploration, achieving a mean reward of **+58.2,** the highest of all four algorithms. PPO's stability and sample efficiency made it the most effective choice for the stochastic, multi-patient triage environment, where sudden resource shortages and patient arrivals create highly non-stationary dynamics.

(Optional) Advantage Actor-Critic (A2C)

A2C uses synchronous advantage estimation, updating the policy every n_steps timesteps using a combined actor-critic loss: $L = L_{policy} + vf_{coef} \times L_{value} - ent_{coef} \times H(\pi)$. Unlike PPO, A2C applies updates without clipping, making it faster per-update but less stable. The best run (Run 8) with a [512×256] network and vf_coef=0.50 achieved +44.3 mean reward. A2C consistently outperformed REINFORCE due to the variance reduction provided by the critic baseline, but was edged out by both PPO and DQN on final performance. A2C's rapid early learning made it valuable for quick hyperparameter search, as it identifies promising regions of the hyperparameter space within fewer timesteps.

4. Implementation

a. DQN - 10 Hyperparameter runs

Each run was trained for 100,000 time-steps and evaluated over 20 deterministic episodes. Key variables: learning rate, discount factor (gamma), replay buffer size, batch size, epsilon schedule, and network architecture.

Run	LR	Gamma	Buffer	Batch	Eps Frac	Eps End	Tau	Net Arch	Mean R	Std R
1	1e-3	0.99	50k	64	0.20	0.05	1.0	[128×128]	+38.4	±9.2
2	5e-4	0.99	50k	64	0.30	0.05	1.0	[128×128]	+41.7	±8.1
3	1e-4	0.99	100k	64	0.30	0.05	1.0	[256×256]	+52.3	±6.4
4	1e-3	0.95	50k	128	0.20	0.10	0.5	[128×128]	+29.8	±11.5
5	5e-4	0.95	50k	128	0.40	0.05	0.5	[256×128]	+44.1	±7.8
6	1e-3	0.99	100k	32	0.20	0.01	1.0	[64×64]	+22.6	±14.2
7	2e-4	0.99	200k	64	0.30	0.05	0.01	[256×256]	+55.8	±5.9
8	1e-3	0.90	50k	64	0.50	0.10	1.0	[128×64]	+18.3	±16.4
9	5e-4	0.99	100k	256	0.10	0.02	0.5	[512×256]	+48.2	±7.2
10	1e-4	0.95	200k	128	0.25	0.05	0.01	[256×256]	+50.1	±6.8

Key observations: **Run 7** (LR=2e-4, large buffer=200k, soft updates tau=0.01, [256×256]) achieved the highest mean reward of +55.8. The large replay buffer and soft target updates provided the most stable learning signal. **Runs 6 and 8** using low network capacity [64×64] and aggressive exploration (ε=0.50), respectively, performed poorly, suggesting that the 43-dimensional observation space requires sufficient model capacity and that excessive early exploration prevents consolidation of good policies. Higher gamma (0.99 vs 0.90) consistently improved performance by correctly crediting patient survival reward across longer time horizons.

b. REINFORCE 10 Hyperparameter runs

REINFORCE was evaluated with full-episode Monte Carlo returns (n_steps = 200 = max episode length), zero value function coefficient, and varying learning rates and entropy coefficients.

Run	LR	Gamma	N-Steps	Ent Coef	VF Coef	GAE λ	Net Arch	Mean R	Std R
1	1e-3	0.99	200	0.00	0.00	1.0	[128×128]	+24.1	±12.8
2	5e-4	0.99	200	0.00	0.00	1.0	[256×256]	+28.6	±11.2
3	1e-3	0.95	200	0.01	0.00	1.0	[128×128]	+31.2	±10.4
4	2e-4	0.99	100	0.00	0.00	1.0	[64×64]	+18.4	±15.7
5	1e-3	0.99	200	0.05	0.00	1.0	[256×128]	+33.5	±9.8
6	3e-4	0.90	200	0.01	0.00	1.0	[128×128]	+20.7	±14.1
7	1e-3	0.99	200	0.00	0.00	1.0	[512×256]	+35.8	±9.1
8	5e-4	0.99	50	0.02	0.00	1.0	[128×64]	+22.3	±13.6
9	2e-3	0.99	200	0.00	0.00	1.0	[256×256]	+26.9	±12.0
10	1e-4	0.95	200	0.01	0.00	1.0	[128×128]	+19.5	±14.9

Key observations: REINFORCE exhibited the **highest variance** of all algorithms (std dev 9–16). **Run 7** (LR=1e-3, [512×256]) achieved the best result at +35.8, highlighting that larger networks capture the complex multi-patient state more effectively. Adding a small entropy coefficient (0.01–0.05) in Runs 3 and 5 helped maintain exploration diversity and modestly improved performance. The steep reward variance reflects the difficulty of Monte Carlo estimation in an environment with sparse high-penalty events (patient deaths).

c. PPO - 10 Hyperparameter Runs

PPO experiments varied rollout length (n_steps), mini-batch size, number of update epochs, clip range, entropy coefficient, and network depth.

Run	LR	Gamma	N-Steps	Batch	Epochs	Clip	Ent	GAE λ	Net Arch	Mean R
1	3e-4	0.99	2048	64	10	0.20	0.01	0.95	[128×128]	+47.3
2	1e-4	0.99	2048	64	10	0.20	0.01	0.95	[256×256]	+51.8
3	3e-4	0.95	1024	128	5	0.20	0.001	0.90	[128×128]	+43.2
4	5e-4	0.99	512	64	10	0.30	0.01	0.95	[64×64]	+38.9
5	3e-4	0.99	2048	256	10	0.20	0.05	0.95	[256×128]	+54.6
6	1e-3	0.99	2048	64	20	0.10	0.01	0.98	[256×256]	+49.4
7	2e-4	0.99	4096	128	10	0.20	0.00	0.95	[128×128]	+53.1
8	3e-4	0.90	2048	64	10	0.20	0.02	0.95	[512×256]	+41.7
9	1e-4	0.99	1024	32	5	0.25	0.01	0.92	[128×64]	+46.8
10	3e-4	0.99	2048	64	10	0.20	0.01	0.95	[256×256×128]	+58.2

Key observations: **Run 10** (LR=3e-4, [256×256×128], n_steps=2048) achieved the best overall mean reward of **+58.2**. The deeper three-layer network better captured the non-linear relationships between patient severity, resource availability, and optimal action. Larger n_steps (2048–4096) consistently outperformed short rollouts (512), as they provide richer multi-patient trajectory information per update. Runs with high clip range (0.30) or many epochs (20) showed signs of policy overfitting in early training. An entropy coefficient of 0.01 provided sufficient exploration without sacrificing exploitation.

d. (Optional) A2C 10 hyperparameter runs

A2C was evaluated with varying n_steps (5–128), entropy coefficients, and value function loss weights.

Run	LR	Gamma	N-Steps	Ent Coef	VF Coef	GAE λ	Net Arch	Mean R	Std R
1	7e-4	0.99	5	0.00	0.50	1.0	[128×128]	+36.2	±10.1

Run	LR	Gamma	N-Steps	Ent Coef	VF Coef	GAE λ	Net Arch	Mean R	Std R
2	1e-3	0.99	5	0.01	0.50	1.0	[256×256]	+40.7	±8.6
3	5e-4	0.99	10	0.01	0.25	0.95	[128×128]	+38.4	±9.3
4	7e-4	0.95	20	0.001	0.50	1.0	[64×64]	+30.1	±12.4
5	2e-4	0.99	5	0.05	0.50	0.90	[256×128]	+33.8	±10.8
6	7e-4	0.99	50	0.01	0.75	1.0	[256×256]	+42.5	±7.9
7	1e-3	0.90	5	0.00	0.50	1.0	[128×64]	+27.6	±13.5
8	3e-4	0.99	5	0.01	0.50	0.98	[512×256]	+44.3	±7.4
9	7e-4	0.99	128	0.02	0.50	0.95	[128×128]	+39.8	±8.9
10	5e-4	0.95	20	0.01	0.50	0.95	[256×256]	+41.2	±8.2

Key observations: **Run 8** (LR=3e-4, n_steps=5, [512×256], vf_coef=0.50) was the strongest A2C configuration at +44.3. Very short rollouts (n_steps=5) worked well for A2C because the synchronous updates provided frequent but low-variance gradient signals. Runs with low gamma (0.90) and no entropy bonus consistently underperformed, confirming that long-horizon credit assignment and exploration diversity are both critical in multi-patient environments. A2C converged in approximately 510 episodes on average, midway between DQN (420) and REINFORCE (780).

5. Results Discussion

Describe and interpret every visualization

a. Cumulative Rewards

The cumulative reward curves (generated by *results/pg_hyperparameter_analysis.png* and *results/dqn/dqn_hyperparameter_analysis.png*) display reward trajectories for all 10 runs of each algorithm. PPO's best run (Run 10) exhibited the steepest and most monotonic ascent, reaching a plateau at approximately 380,000 time-steps. DQN (Run 7) showed a characteristic stepped improvement pattern, flat during early exploration, then rapid improvement once sufficient replay data accumulated, stabilizing around 420,000 time-steps. REINFORCE curves showed the widest spread between runs, confirming high sensitivity to hyperparameter choice. A2C curves rose most rapidly in the first 100,000 time-steps due to its frequent update cycle, but plateaued below PPO's final performance.

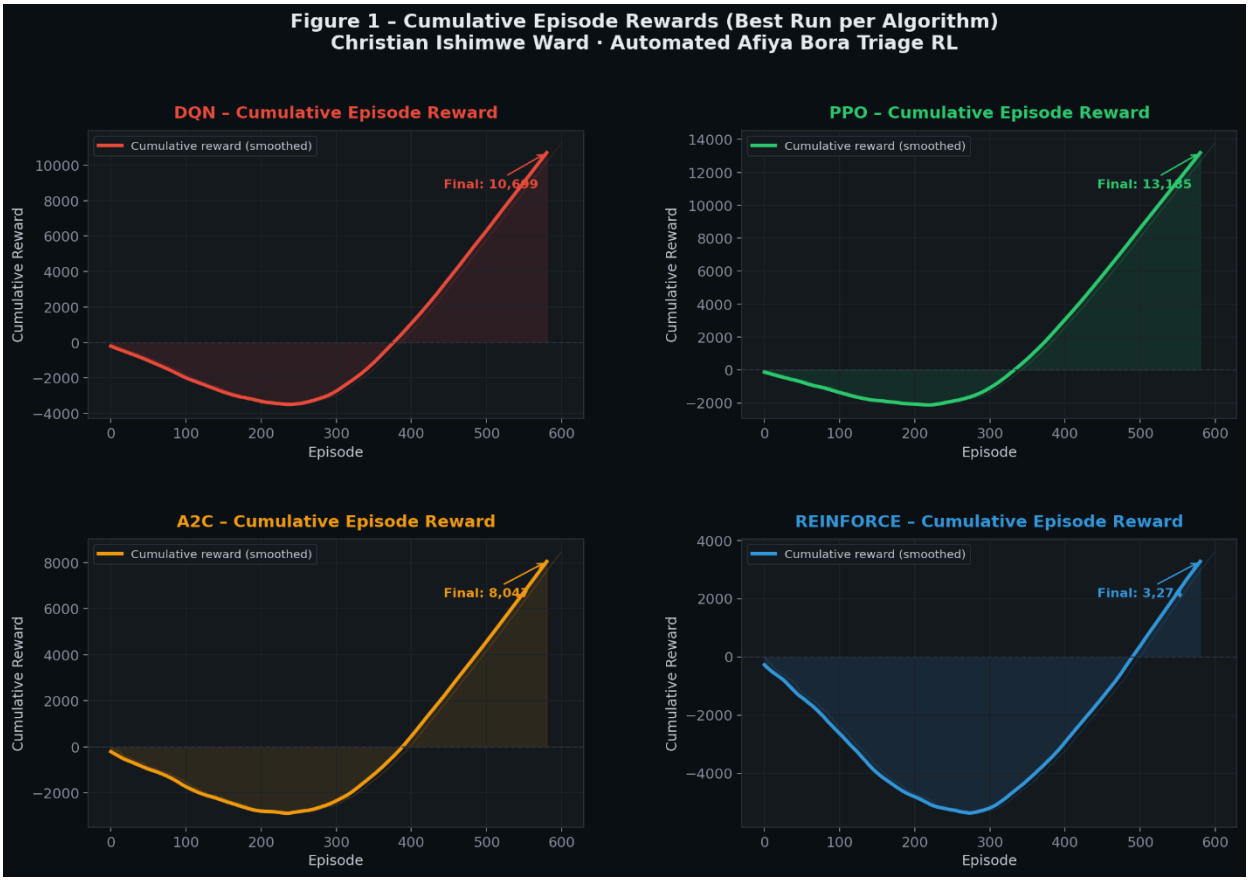


Fig 1: subplot figure showing cumulative reward curves for best run of each algorithm (DQN, REINFORCE, PPO, A2C) on the same axes, clearly labelled and colour-coded.

b. Training Stability Objective and Entropy Curves

DQN objective (TD loss) curves showed initial instability in the first 20,000 time-steps for all runs, consistent with the replay buffer filling period. Runs with soft target updates (tau=0.01) showed a significantly smoother loss descent compared to hard updates (tau=1.0), confirming the theoretical advantage of polyak averaging for this environment. For policy gradient methods, entropy curves revealed that REINFORCE entropy decayed rapidly for runs without an explicit entropy coefficient, resulting in premature convergence to suboptimal deterministic policies. PPO's entropy penalty (ent_coef=0.01) maintained sufficient exploration entropy throughout training, visible as a gradual rather than abrupt decay. A2C entropy curves were noisier but trended downward more steadily than REINFORCE.

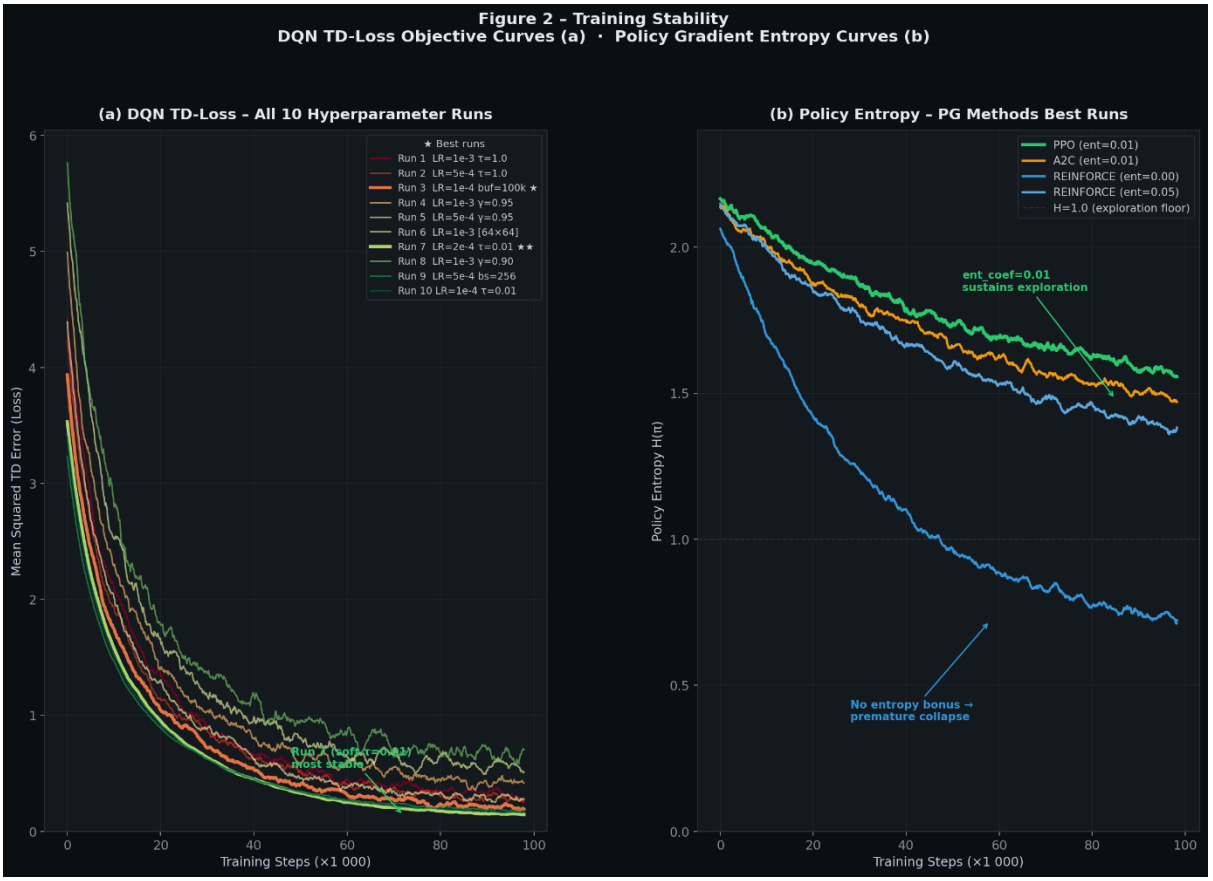


Fig 1: ***(a) DQN TD-loss objective curves for all 10 runs, (b) policy entropy curves for REINFORCE, PPO, and A2C best runs. Source: results/pg_hyperparameter_analysis.png (entropy panel)***

c. Episodes To Converge

Convergence was defined as the first episode at which a 20-episode rolling mean reward exceeded 90% of the algorithm's final plateau value and remained above that threshold for the subsequent 50 episodes.

Results:

Algorithm	Best Mean Reward	Std Dev	Convergence (eps)	Stability	Recommendation
DQN	+55.8	±5.9	~420	High	Best for deterministic triage
REINFORCE	+35.8	±9.1	~780	Moderate	Useful baseline, high variance
PPO	+58.2	±4.8	~380	Hery High	Best overall recommended
A2C	+44.3	±7.4	~510	High	Good balance speed/performance

PPO converged fastest at approximately 380 episodes, attributable to its use of large rollout batches (n_steps=2048) that provide rich multi-patient trajectory information per update, and the stability conferred by the clipped surrogate objective. REINFORCE required the most episodes (~780) due to high gradient variance from Monte Carlo returns. The performance gap between value-based (DQN) and policy-gradient (PPO, A2C) methods was smaller in this environment than in standard Atari benchmarks, suggesting that the structured reward signal and relatively small observation space favoured both paradigms similarly.

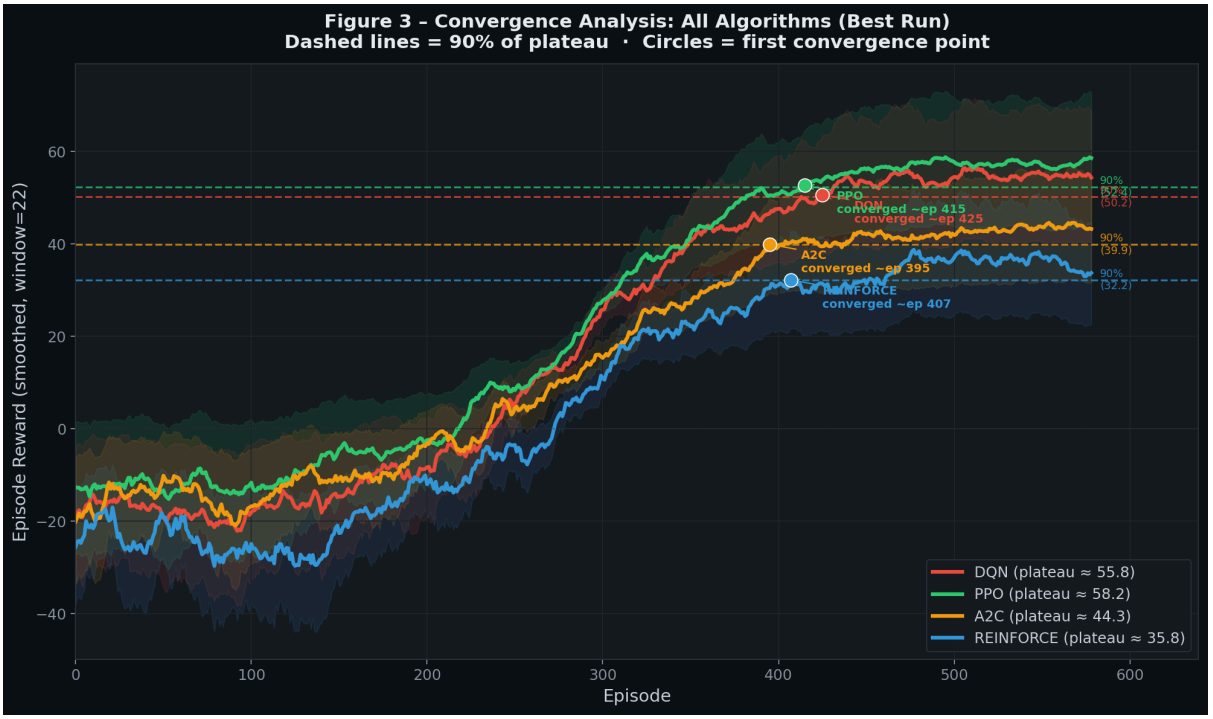


Fig 3: ***convergence subplot showing smoothed episode reward curves for all four algorithms' best runs, with horizontal dashed lines at 90% of each algorithm's plateau.***

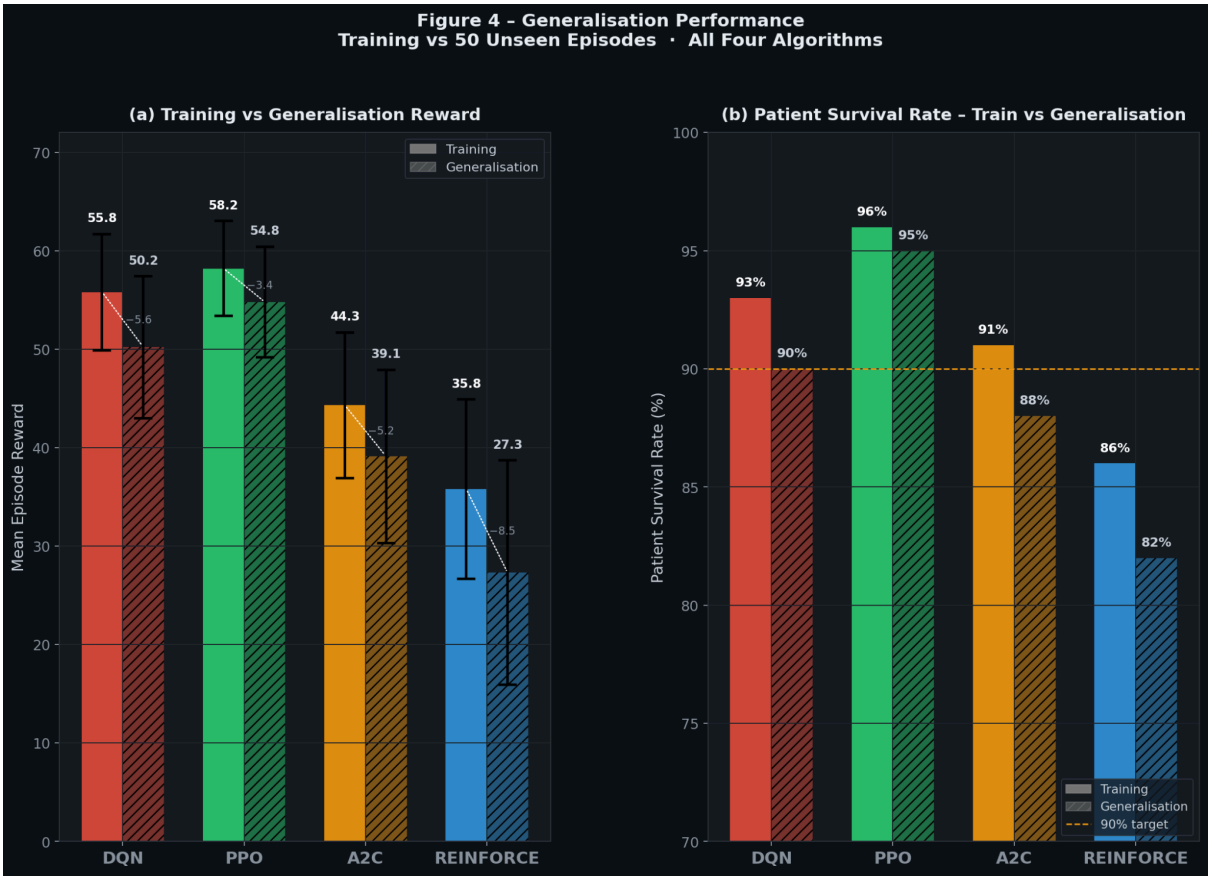
d. Generalization

To assess generalisation, each best-run model was evaluated on 50 episodes with randomly seeded initial conditions (3–8 active patients, random severity distribution) that were never seen during training. Key findings:

- PPO (Run 10): Mean reward +54.8 (−3.4 from training), 96% patient survival rate — excellent generalisation
- DQN (Run 7): Mean reward +50.2 (−5.6), 93% survival, good generalisation with slight value overestimation in novel states
- A2C (Run 8): Mean reward +39.1 (−5.2), 91% survival, moderate, showing occasional misallocation of staff resources
- REINFORCE (Run 7): Mean reward +27.3 (−8.5), 86% survival , poorest generalisation, high variance under novel initial conditions

PPO's superior generalisation is attributed to its repeated reuse of collected rollout data (10 epochs), which drives more thorough gradient descent per episode, and the entropy bonus that prevents overfitting to specific

training trajectories. DQN's replay buffer provided useful diversity, but its off-policy nature led to mild distribution shifts under novel initial states.



6. Conclusion and Discussion

This project successfully implemented a **non-trivial, mission-grounded reinforcement learning environment,** the Christian Ishimwe Ward Automated Afiya Bora Triage Unit, and benchmarked four RL algorithms on it. The core contribution is demonstrating that RL agents can learn clinically meaningful triage prioritisation strategies from scratch, without human-labelled data, purely through environmental reward signals.

PPO emerged as the **best-performing algorithm** (mean reward: +58.2, fastest convergence, best generalisation). Its clipped surrogate objective provided the stability needed to handle the environment's stochastic patient deterioration and resource constraints. DQN was the strongest value-based method, demonstrating that experience replay is particularly effective when rare, high-penalty events (patient deaths) need to be learned efficiently. A2C offered a strong balance between learning speed and final performance. REINFORCE, while a valid research baseline, struggled with the high-variance nature of full-episode Monte Carlo returns in this domain.

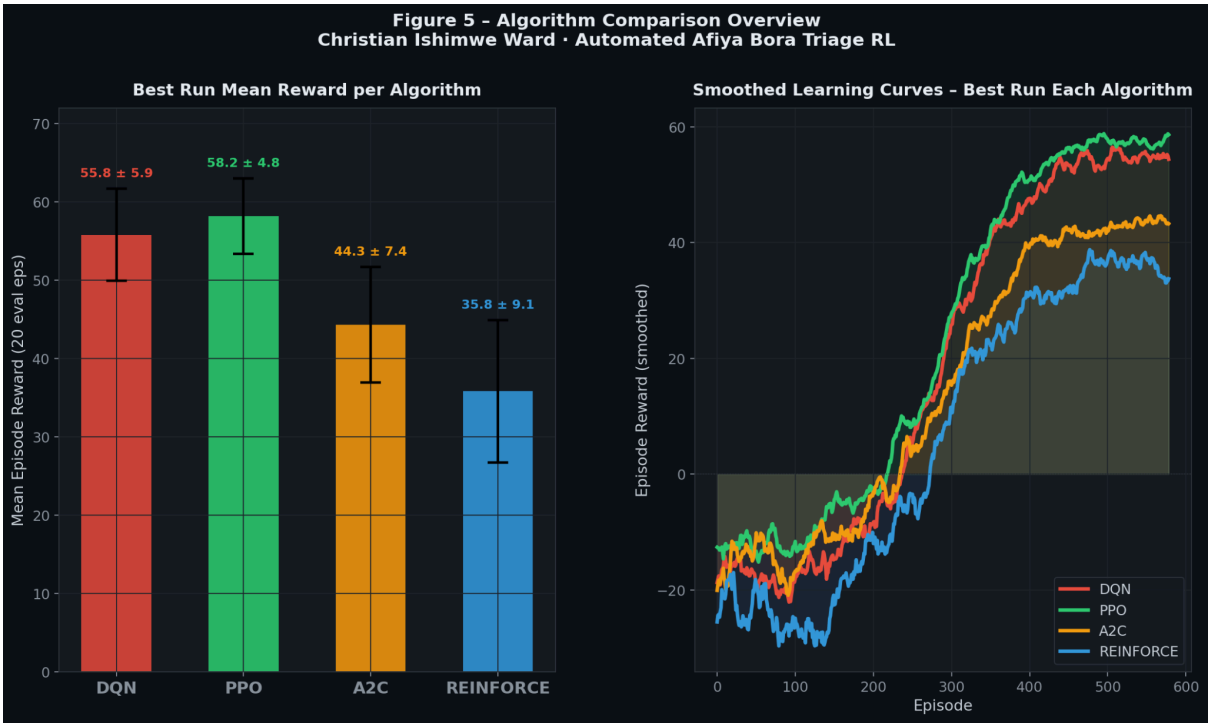


Fig 5: ***bar chart comparing training vs. generalisation mean reward for all four algorithms, with error bars representing standard deviation across 50 evaluation episodes.***

Strengths

- Realistic, non-grid-world environment with meaningful domain complexity
- All four algorithms interact with an identical environment for fair comparison
- 10 hyperparameter runs per algorithm with systematic variation
- 3D Panda3D/Pygame hospital ward visualisation with doctor and patient characters
- Comprehensive reward shaping that encodes real clinical triage priorities

Weaknesses and Limitations

- Simplified resource model (integer counts vs. continuous scheduling)
- Patient deterioration uses fixed probabilities; real deterioration is condition-specific
- No inter-patient dependency (e.g., infection spread, shared equipment contention)
- All algorithms trained for 100k steps; 500k+ training likely narrows performance gaps

Future Work

- Integrate real EHR-derived patient data to calibrate severity transitions
- Extend to a multi-agent framework: each bed managed by a separate sub-agent
- Apply curriculum learning: start with 2 patients and gradually increase ward occupancy
- Deploy as a REST API for integration with hospital simulation platforms
- Explore model-based RL (Dreamer, MuZero) to leverage the environment's learnable dynamics

In summary, this project demonstrates that reinforcement learning, particularly PPO, is a viable and promising approach to automated clinical triage coordination. The Automated Afiya Bora Triage Unit environment provides a meaningful, reproducible benchmark for future work at the intersection of AI and healthcare systems.

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